

Name: _____



Converting terminating decimals to fractions

Task A: Fractions less than 1

1) Convert the following into simplified fractions:

a) 0.65

b) 0.34

c) 0.12

d) 0.542

e) 0.864

f) 0.1248

2) Laura and Jacob are given 0.56 and 0.560 to convert into a fraction.



Laura
says

The answers will be different as 0.56 is two decimal places and 0.560 is 3 decimal places.

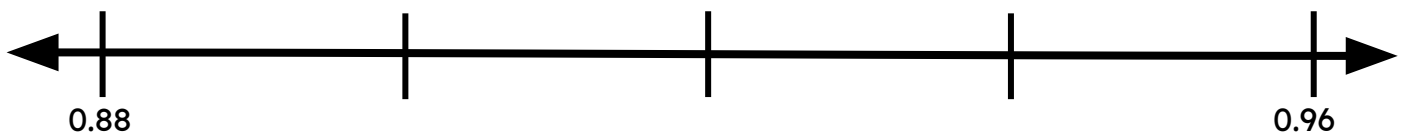


Jacob
says

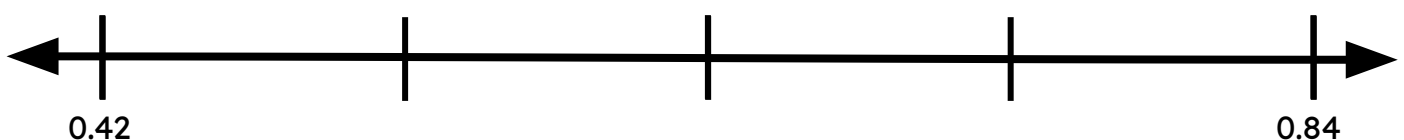
It is the same decimal so it will be the same fraction.

Who is correct? Work out the fractional equivalent to the decimal(s)

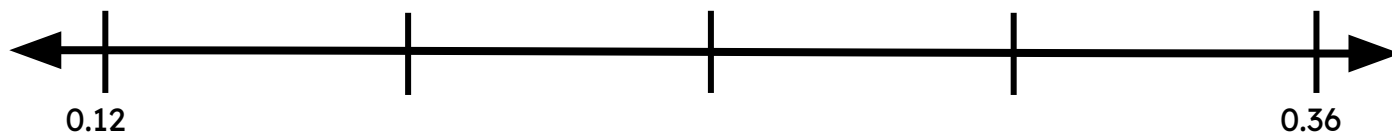
3a) Write the fraction which is halfway between 0.96 and 0.88



3b) Write the fraction which is a quarter of the way between 0.42 and 0.84



Three students created their own method to find the middle fraction.



3c) Do all three methods work? Explain.



Izzy

I converted 0.12 and 0.36 into fractions. You can see the middle easily.



Aisha

Add 0.36 with 0.12, then halve the answer. Then find the fraction of the decimal.



Alex

Work out the difference between 0.36 and 0.12, then halve it and add it to 0.12. Convert this to a fraction.

Task B: Fractions greater than 1

1) Convert the following decimals into mixed numbers and improper fractions.

Decimal	Mixed number	Improper fraction
2.45		
8.125		
8.288		
9.72		



2)

π is a special symbol and represents the non-terminating decimal 3.141592654...

There is no fractional equivalent to the exact value of π .

Work out the mixed number equivalent to:

- a) 3.141
- b) 3.1415
- c) 3.141592

Task C: Writing the denominator in exponential form

1) Work out the power of the exponent.

- a) $0.4567 = \frac{4567}{10^{\square}}$
- b) $0.00017 = \frac{17}{10^{\square}}$
- c) $3.001 = \frac{3001}{10^{\square}}$

2) Write the following decimals as a fraction where the denominator is in exponential form.

- a) 0.1257
- b) 3.567
- c) 0.00023

a) $0.167 = \frac{167}{1000} = \frac{167}{10^3}$

b) $0.0357 = \frac{357}{10000} = \frac{357}{10^4}$

c) $0.15 = \frac{15}{100} = \frac{15}{100}$

d) $\frac{98}{125} = \frac{98}{1000} = \frac{98}{10^3}$